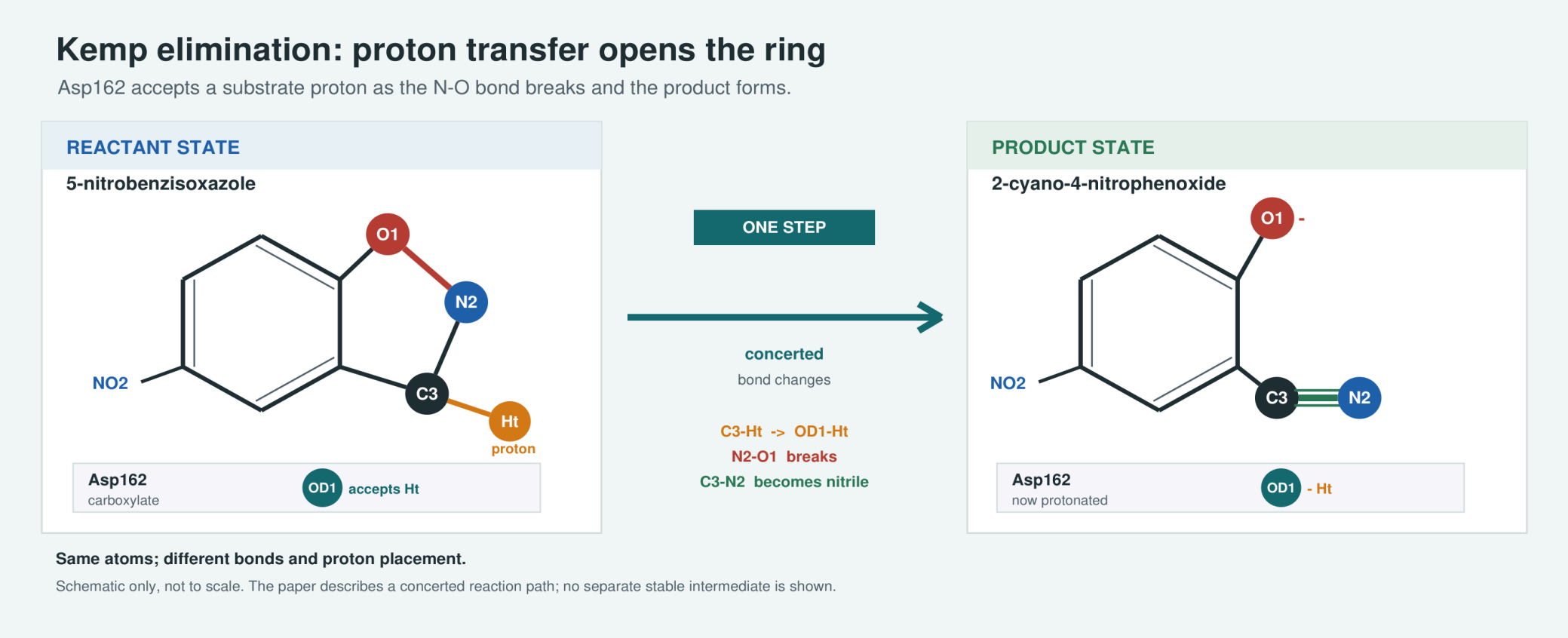
REACTION AT A GLANCE

# Kemp elimination: what changes?

Asp162 accepts a proton from 5-nitrobenzisoxazole as the substrate ring opens to form the cyanophenoxide product.



UNDERSTANDING EVB

# Two bonding descriptions, one reaction

REACTANT STATE (R)

## Before

The substrate still has its C-H bond and closed N-O-containing ring. Asp162 is a carboxylate ready to accept the proton.

PRODUCT STATE (P)

## After

Asp162 has accepted the proton. The N-O bond is broken, the nitrile has formed, and the ring-opened phenoxide product remains.

EVB

## How the states mix

EVB combines energy descriptions for R and P. The energy offset shifts one state relative to the other; coupling controls how strongly they mix.

## Explore the saved examples in this order

| | | |
|---|---|---|
| 1 | **EVB fundamentals** | Change coupling and offset one at a time. |
| 2 | **Saved reaction profile** | Find the reactant, high-energy region, and product. |
| 3 | **Parameter comparison** | Compare parameter pairs and their product energies. |

Paper Figure 3e shows barrier estimates; the OpenMM viewer shows a free-energy profile. Different boundaries, electrostatics, Morse references, and sampling mean these are not a direct comparison.

FULL CALCULATION WORKFLOW

# From a structure to a free-energy profile

1 · PREPARE

## Build matched R and P

Use the same atoms in the same order. Assign state-specific bonds, charges, force-field terms, and EVB settings. Check the atom map, total charge, proton position, solvent, and ions before running.

2 · RELAX

## Stabilize both endpoints

Minimize and equilibrate R and P separately. Inspect geometry, contacts, and energy stability. Relaxation prepares each endpoint; it does not calculate the reaction barrier.

3 · FEP/PROFILE

## Sample the lambda path

Sample a set of lambda windows between the state descriptions. Check neighboring-window overlap and convergence, compare independent replicas, then analyze the barrier and reaction free energy.

## What this workshop contains

This repository includes saved Kemp data and browser exercises for learning the method. It does not include the complete live R/P builder, endpoint-relaxation inputs, or lambda-window run scripts, so the three simulation stages above are explained but not launched here.